Barrier function Method for Constrained Optimization

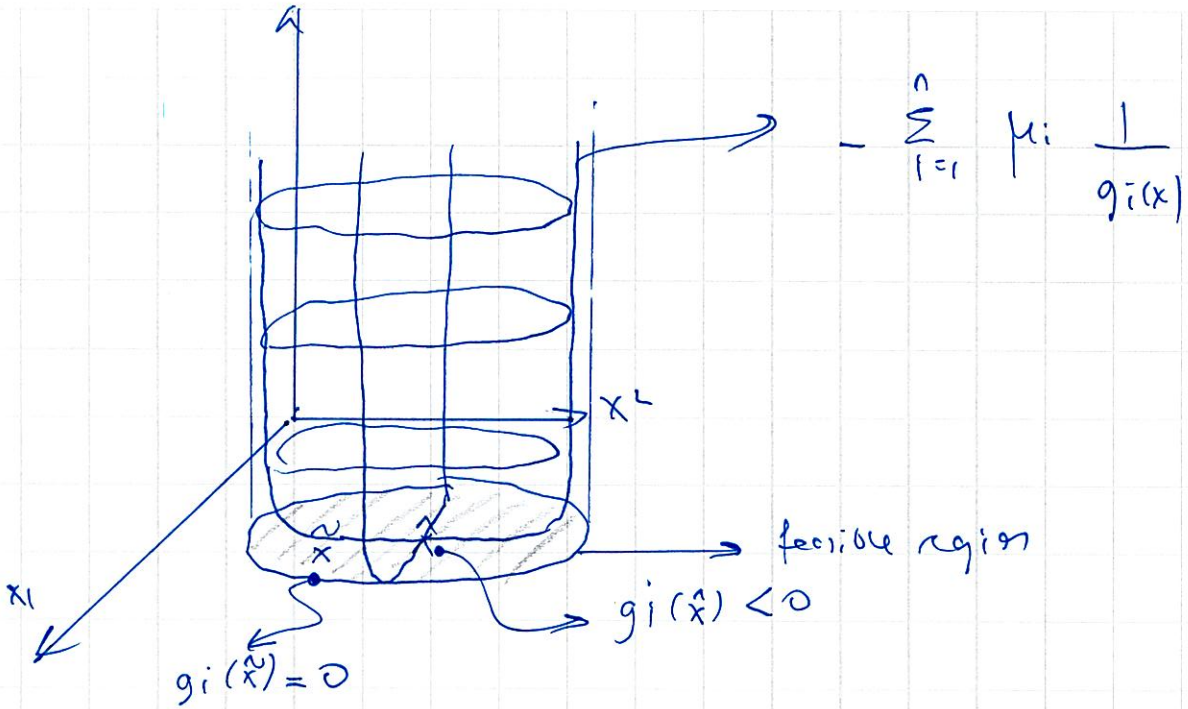
Applicable to problems with only inequality constraints

$$\min f(x)$$
$$g_1(x) \le 0$$
$$\vdots$$
$$g_n(x) \le 0$$

- when we are close to the boundary of first constraints within inside, $g_1(x)$ is a very small, negative number
- so $-g_1(x)$ is a very small, positive number
- Thus, $1/-g_1(x)$ is a very large positive number.
- Solve the problem:

$$\min_x f(x) - \sum_{i=1}^{n} \mu_i \frac{1}{g_i(x)},$$

where $\mu_1, \dots, \mu_n$ are small barrier multipliers


$-\sum_{i=1}^{n} \mu_i \frac{1}{g_i(x)}$
x^2
x_1
$\tilde{x}$
$\hat{x}$
feasible region
$g_i(\hat{x}) < 0$
$g_i(\tilde{x}) = 0$

1. Choose the initial barrier parameters $\mu_1^1, \dots, \mu_n^1$, pick a stopping tolerance ε. pick a shrinkage factor $\beta \in (0,1)$
   Set iteration counter $k=1$
   Pick x^0 strictly inside feasible region, $g_i(x^0) < 0$ $i=1,\dots,n$

2. Solve the problem

$$x^k = \arg\min f(x) - \sum_{i=1}^{n} \mu_i^k \frac{1}{g_i(x)}$$

by using gradient descent by starting from the initial point x^{k-1}

3. If $-\sum_{i=1}^{n} \mu_i^k \frac{1}{g_i(x^k)} \leq \varepsilon$ then stop

else $\mu_i^{k+1} = \beta \cdot \mu_i^k$ $i=1,\dots,n$, increase iteration counter by 1 and go to step 2

- $\min \ (x_1-2)^4 + (x_1-2x_2)^2$ st $x_1^2 - x_2 \leq 0$

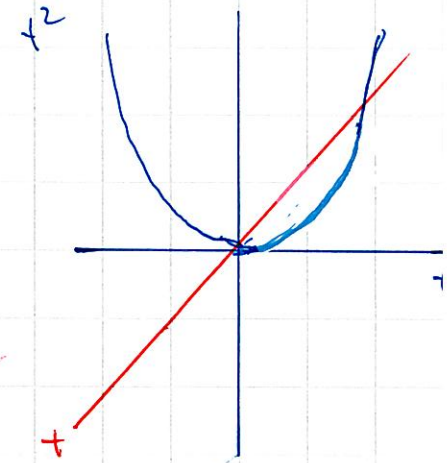
Taylor's Theorem:

We will use the notation $o(t)$ to denote any single-dimensional function $g: \mathbb{R} \to \mathbb{R}$ that satisfies

$$\lim_{t \to 0} \frac{g(t)}{t} = 0$$

- For example, $g(t) = t^2$ is $g(t)$ $o(t)$ function?

$$\lim_{t \to 0} \frac{g(t)}{t} = \lim_{t \to 0} \frac{t^2}{t} = \lim_{t \to 0} t = 0$$ Yes

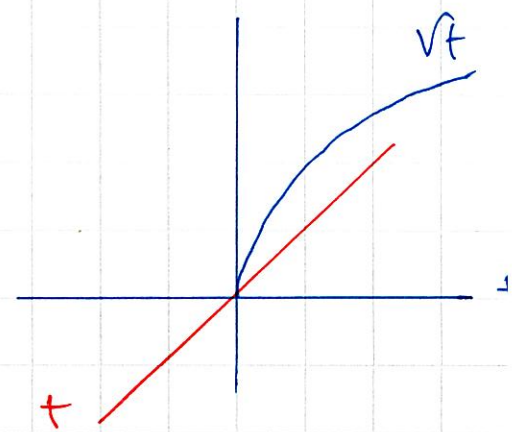


- For example, $g(t) = 2t$ is $g(t)$ $o(t)$ function? No

$$\lim_{t \to 0} \frac{g(t)}{t} = \lim_{t \to 0} \frac{2t}{t} = 2$$

- For example $g(t) = \sqrt{t}$ is $g(t)$ $o(t)$ function? No

$$\lim_{t \to 0} \frac{g(t)}{t} = \lim_{t \to 0} \frac{\sqrt{t}}{t} = \lim_{t \to 0} \frac{1}{\sqrt{t}} = \infty$$

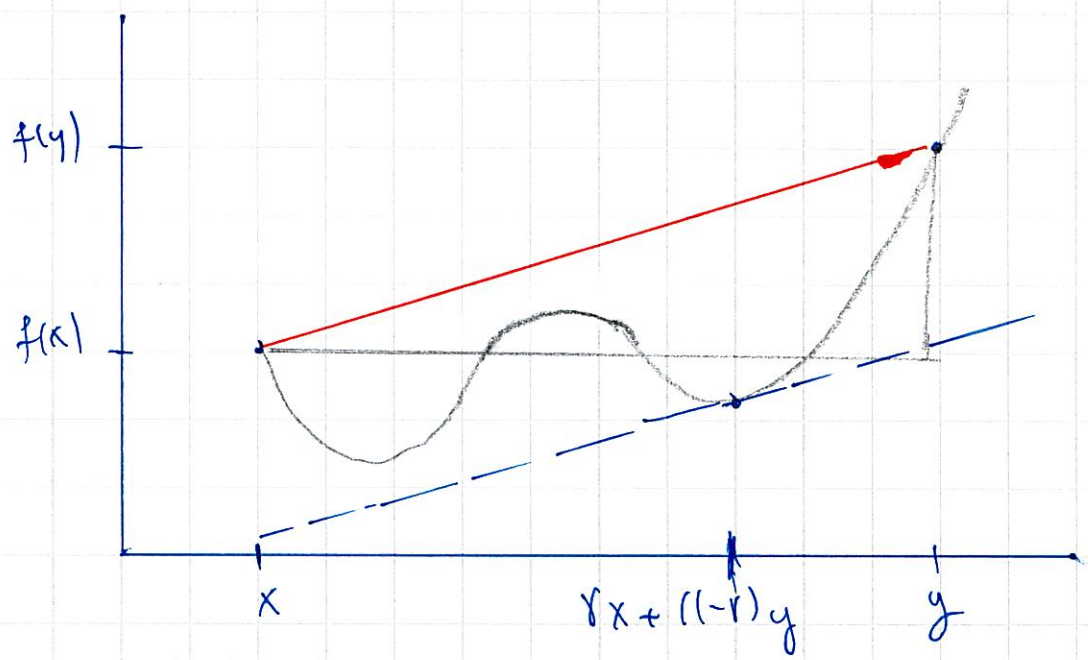


- Let's focus on infinitely differentiable function $f: \mathbb{R}^n \to \mathbb{R}$. This means that

$$\frac{\partial^k f(x)}{\partial x_{i_1} \partial x_{i_2} \dots \partial x_{i_k}} \quad \text{exists for all} \quad k = 1, 2, \dots$$

- Taylor's Theorem:

$$f(y) = f(x) + \nabla^T f(\gamma x + (1-\gamma) y) \cdot (y - x) \quad \text{for some} \quad \gamma \in (0,1)$$



- Recall $\|x\| = \sqrt{\sum_{i=1}^{n} x_i^2}$ $\qquad x^t \cdot y = \|x\| \, \|y\| \cdot \cos\theta$, where θ is the angle between x and y

- Implication of Taylor's Theorem:

$$f(y) = f(x) + \nabla^t f(x) \cdot (y-x) + o(\|y-x\|)$$

Pf: By taylor's thm: $f(y) = f(x) + \nabla^t f(\gamma x + (1-\gamma) y) \cdot (y-x)$ for some $\gamma \in (0,1)$

$$f(y) = f(x) + \nabla^t f(\gamma x + (1-\gamma) y)\,(y-x)$$

$$= f(x) + \nabla^t f(x)\,(y-x) + \left(\nabla^t f(\gamma x + (1-\gamma) y) - \nabla^t f(x)\right)(y-x)$$

$$= f(x) + \nabla^t f(x)\,(y-x) + \|\nabla^t f(\gamma x + (1-\gamma) y) - \nabla^t f(x)\| \, \|y-x\| \cos\theta$$

where θ is the angel between $\nabla^t f(\gamma x + (1-\gamma) y) - \nabla^t f(x)$ and $y-x$

If we can show that $\|\nabla^t f(\gamma x + (1-\gamma) y) - \nabla^t f(x)\| \, \|y-x\| \cos\theta$ is $o(\|y-x\|)$ then we are done.

Check: $\lim_{\|y-x\| \to 0} \dfrac{\|D^t f(\gamma x + (1-\gamma) y) - D^t f(x)\| \cancel{\|y-x\|} \, C_0, \theta}{\cancel{\|y-x\|}} = 0$

because if $\|y-x\| \to 0$, then $y - x \to 0$ then

$D^t f(\gamma x + (1-\gamma) y) - D^t f(x) \to 0$